

Experimental Design for In Vitro Colon Models

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Third INFOGUT Training School - September 28-30, 2026. Madrid, Spain
Bioinformatics 101 for colon models experiments



Design failures are invisible, until someone tries to replicate you

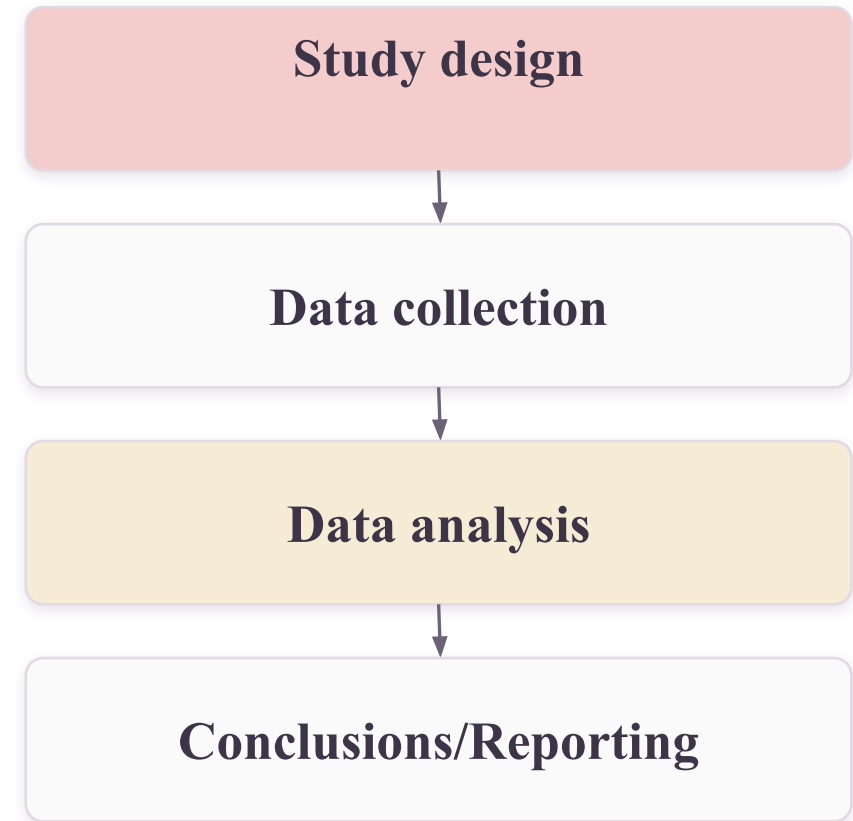
“Across biology, most irreproducible findings trace back to how the experiment was designed, not which statistical test was run afterward.”

Session overview

- Sources of variation: donor, vessel, batch/run, and technical
- Biological vs. technical replicates (pseudoreplication)
- Controls (blank, vehicle, positive, baseline)
- Confounders and randomization/blocking
- Longitudinal sampling and autocorrelation
- Statistical power and sample-size calculation
- Paired/baseline-corrected designs

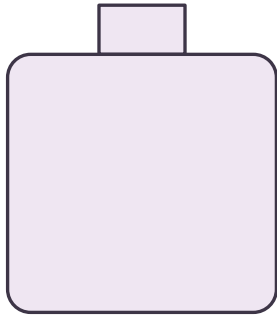
Before you trust a p-value, trust the design

- 1 to 3 vessels per condition is still common in published in vitro colon studies, often too few to detect anything but the largest effects.
- Small studies don't just miss real effects, they systematically inflate the size of the ones they do find.



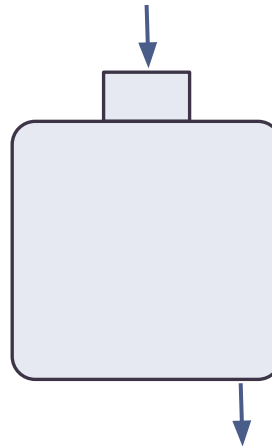
A quick map of the systems

Every design choice below applies to any of these platforms. The details differ, the variance sources don't.



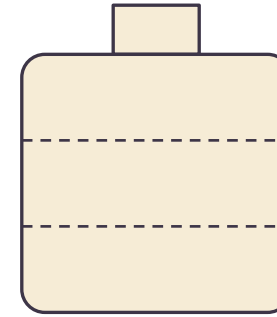
Batch fermentation

Sealed vessel, single inoculum, hours-long run.



Continuous systems

Chemostats, SHIME, TIM-2, M-ARCOL: continuous feed and effluent removal. Steady state over days to weeks.

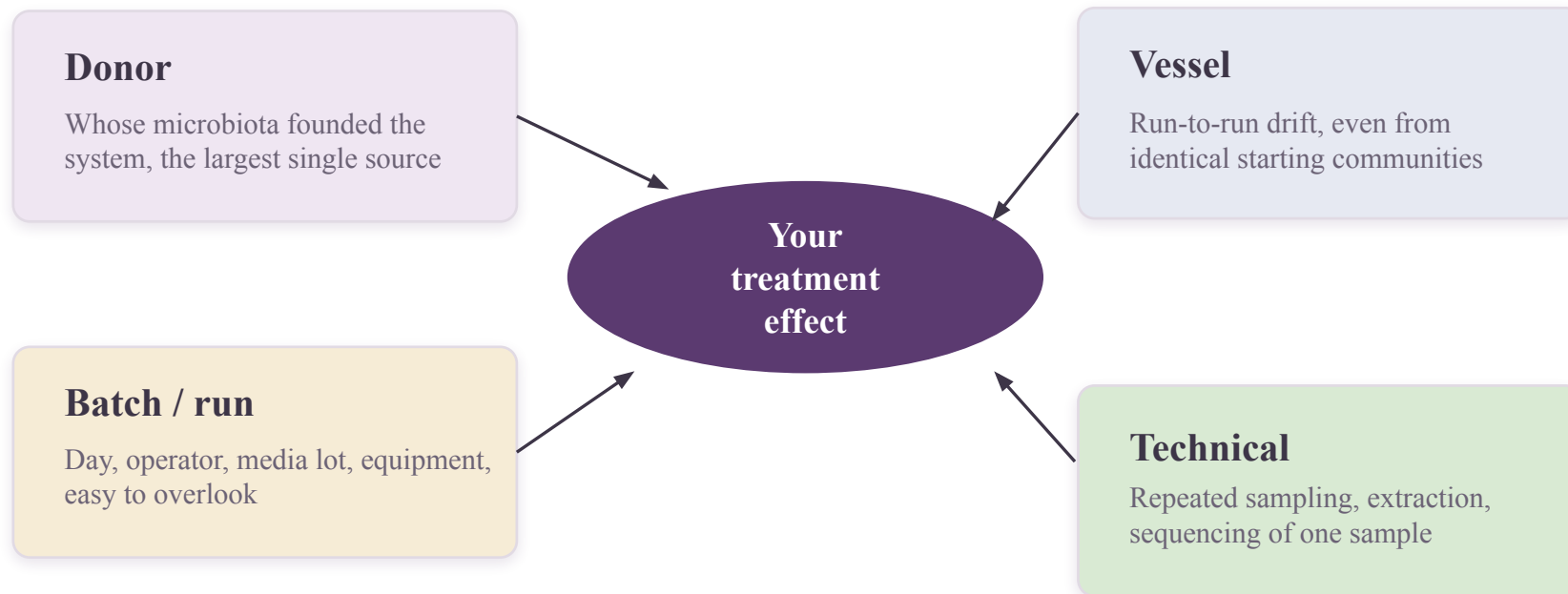


Multi-compartment & mucosal

Simulated colon regions with luminal and mucosal phases (e.g. M-SHIME).

Four sources of variation compete with your effect

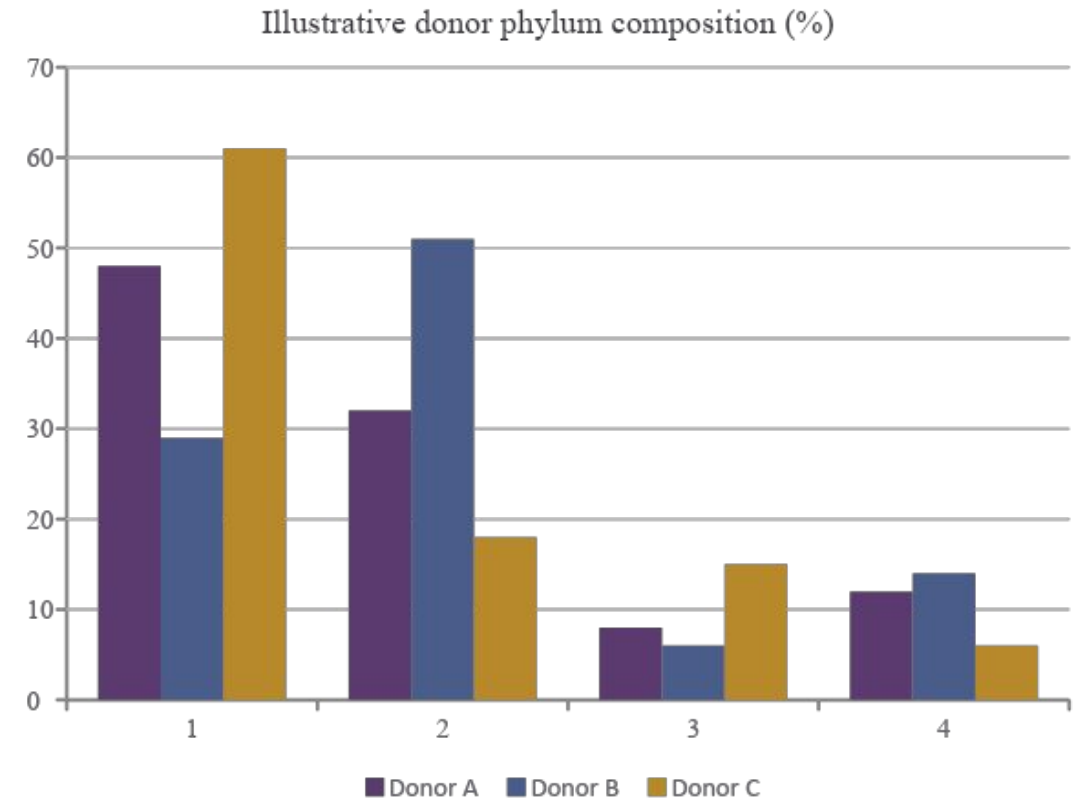
Each box below is a real, estimable source of variance, something your design must isolate.



Your experiment must be designed to separate these four from the effect you actually care about — that separation is what the rest of this lecture builds.

Inter-individual variability is the largest source of variance

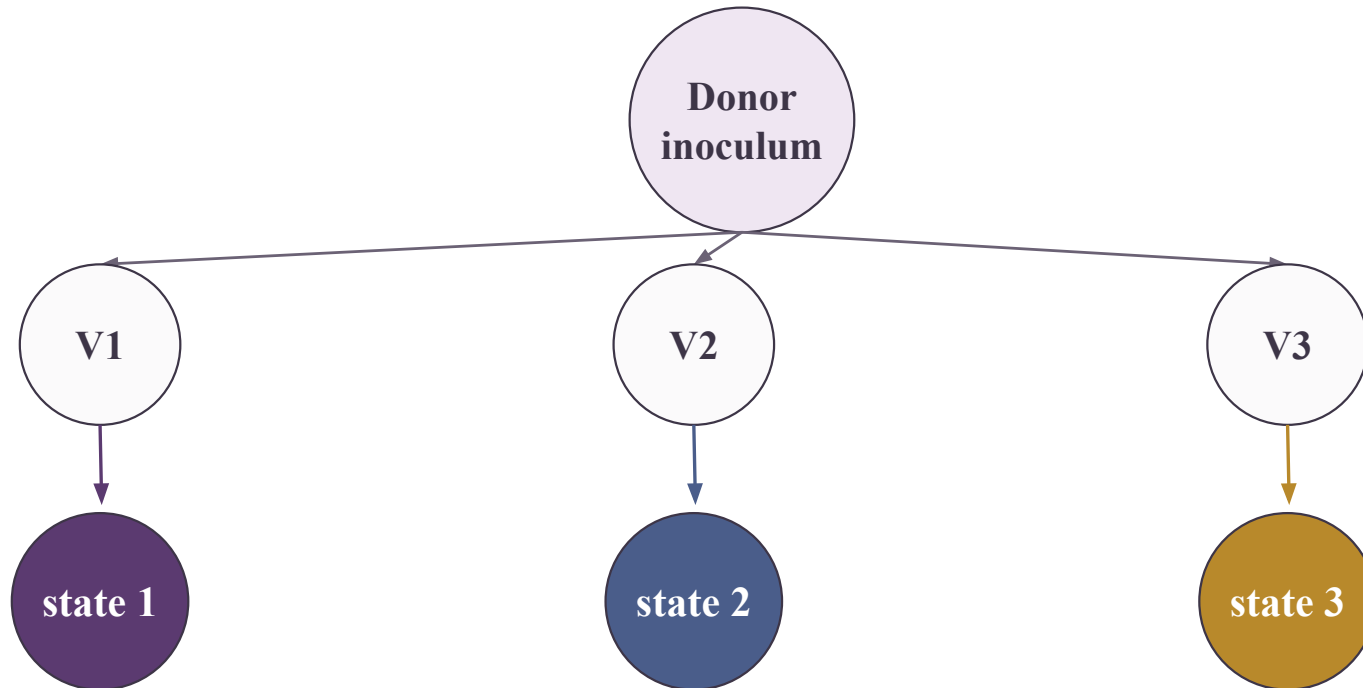
- Composition, diversity, and resilience differ enormously from one donor to the next, often more than any treatment you'll apply.
- A single-donor pilot tells you about that donor's microbiota, not “the microbiome”.
- Treat donor as a blocking / random-effect factor to be modeled, not a nuisance to average away.
- More donors improves generalizability; fewer donors buys more vessels per donor.



Illustrative values, not real data . The spread matters, not the exact numbers (Knight et al., 2018, Nat Rev Microbiol).

Identical inputs, divergent communities

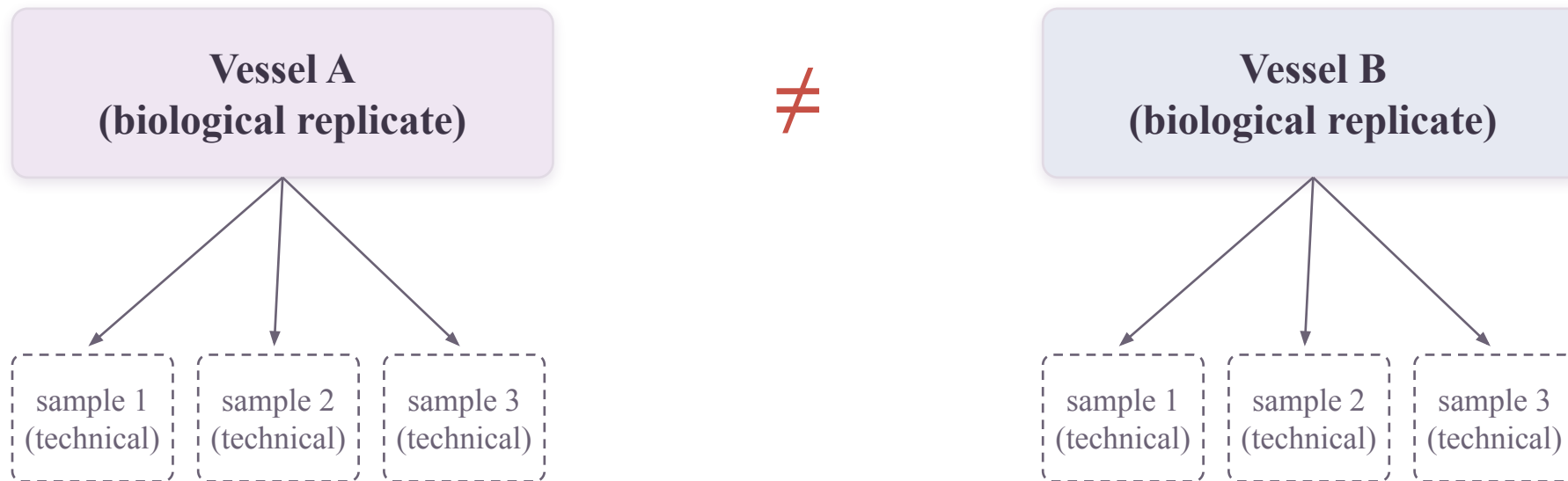
Split one donor's inoculum into parallel vessels and, days later, each has settled into its own slightly different community.



Parallel vessels from the same donor still drift apart via stochastic colonization and selection . Vessel is a real source of variance, not just measurement noise, and it's why “one vessel per condition” is not a real experiment.

Know what your 'n' actually counts

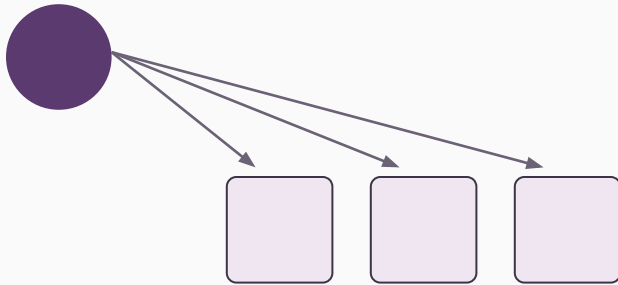
This single distinction causes more inflated significance in in vitro microbiome papers than any other design mistake.



- Biological replicate is an independent vessel or donor, with its own founding community.
- Technical replicate is a repeated sampling, (sequencing from the same vessel), and measures the same biological replicate.
- Treating three technical replicates as three biological ones triples your apparent n and shrinks your p-value.

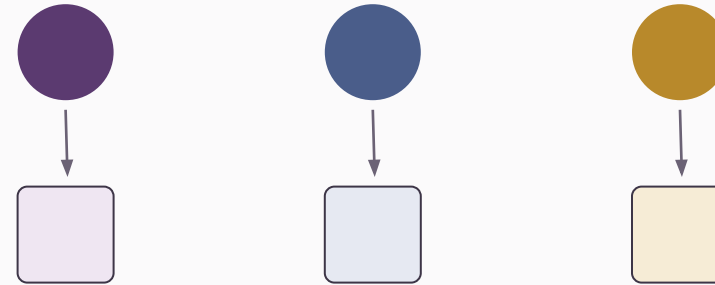
How many vessels, how many donors?

Within-donor replication



One donor, several parallel vessels. Controls for donor variability by holding it constant, but conclusions only generalize to that one donor's microbiota.

Across-donor replication



One vessel per donor, several donors. Costs more vessels for the same total n , but the result generalizes across the population you sampled donors from.

Best practice where feasible: donors $\times$ vessels-per-donor, a nested design analyzed with donor as a random effect. Typical published studies: 3–6 donors, 2–3 vessels per condition, a floor, not a target.

Every treatment needs a matched comparator

Keep the baseline control in mind, it becomes the key to a much cheaper sample size calculation later in this lecture.



Blank / untreated

Tracks natural drift over the incubation, independent of any treatment



Vehicle control

Isolates the compound's effect from its solvent or delivery matrix



Positive control

A compound with a known effect, confirms the system responds at all



Pre-treatment baseline

Same vessel, sampled before dosing, so it serves as its own reference

The variables you didn't mean to test

- Media batch, incubator drift, operator technique, anything that changes between sessions.
- Reactor position or feeding order within a run, detail that can still track your grouping.
- Time-of-harvest and processing delay before extraction, which shift community composition on their own.
- Sequencing batch or lane effects downstream of the wet lab, invisible until you check for them.

Treatment assignment

Control	Control	Control	Treated	Treated	Treated
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Operator (hidden confound)

Op. A	Op. A	Op. A	Op. B	Op. B	Op. B
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Confounded: operator changes exactly track treatment

Randomize, block, and run in parallel

- Randomize vessel position, feeding order, and processing order so hidden variables scatter evenly across groups.
- Block by donor and by run. Compare within blocks, not across them, so the block itself absorbs that variance.
- Run every condition within a batch in parallel: same day, same reagent lot, same operator rotation.
- Record run metadata even if you don't plan to model it yet, you may need it later to explain an outlier.

Treatment assignment

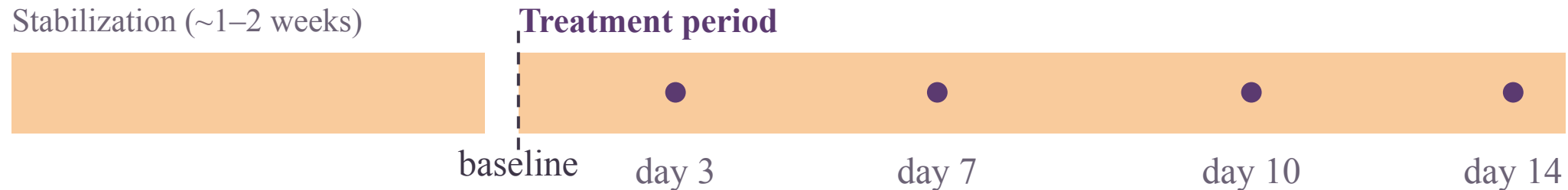
Control	Treated	Control	Treated	Control	Treated
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Operator (no longer confounded)

Op. A	Op. B	Op. A	Op. B	Op. A	Op. B
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Balanced: operator no longer tracks treatment

Repeated samples from one vessel aren't independent



- Samples from the same vessel over time are autocorrelated, so four timepoints are not four independent observations.
- Define “baseline” per vessel, not pooled across vessels, because vessels stabilize to different states even under identical protocols.
- This is exactly what repeated-measures and mixed-effects models exist to handle, by modeling vessel as a random effect (core stats lecture).

A quick gut check

One minute, no need to answer out loud.



- *Which of your last three experiments had a hidden confound you didn't check for?*
- *Right now, could you say whether your reported “n” counts vessels, or samples from one vessel?*
- *If a reviewer asked for your power calculation today, would you have one to show them?*

Turning an effect size into a number of vessels

Given the effect you're hoping to see, how many independent vessels does it take to detect it?

- Statistical power is the probability of detecting a true effect, if it really exists, at your chosen significance threshold.
- It depends on three things you choose before running the experiment:
 - Effect size you care about (d)
 - Significance level (α , conventionally 0.05)
 - Power ($1-\beta$, conventionally 80%)
- The effect size comes from pilot data or comparable published work, never from wishful thinking.

$$n \text{ per group} \approx 2 (z_{1-\alpha/2} + z_{1-\beta})^2 / d^2$$

Two independent groups, comparing means, d is Cohen's d , the standardized difference between group means

$$\alpha = 0.05 \text{ (two-sided)} \quad z_{1-\alpha/2} = 1.96$$

$$\text{Power} = 80\% \quad z_{1-\beta} = 0.84$$

$$\text{Combined constant} \quad 2 \times (1.96 + 0.84)^2 = 15.7$$

Sizing a butyrate comparison

Pilot data: control butyrate 12.0 mM, prebiotic target 16.0 mM, pooled vessel-to-vessel SD 4.0 mM.

1. Standardize the effect

$$d = (16.0 - 12.0) / 4.0 = 1.0$$

A difference of one, a large effect by benchmarks.

2. Plug into the formula

$$n \approx 15.7 / 1.0^2 \approx 15.7 \rightarrow 16$$

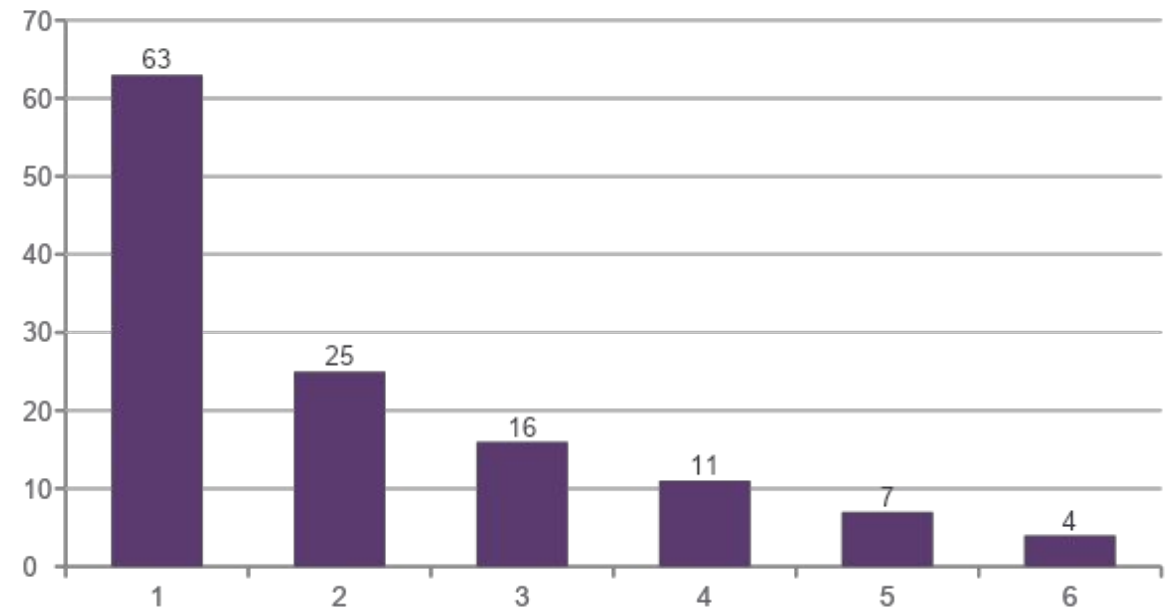
independent vessels per group.

3. Total design size

$$16 \times 2 \text{ groups} = 32 \text{ vessels}$$

The full unpaired, two-group comparison needs 32 independent biological units.

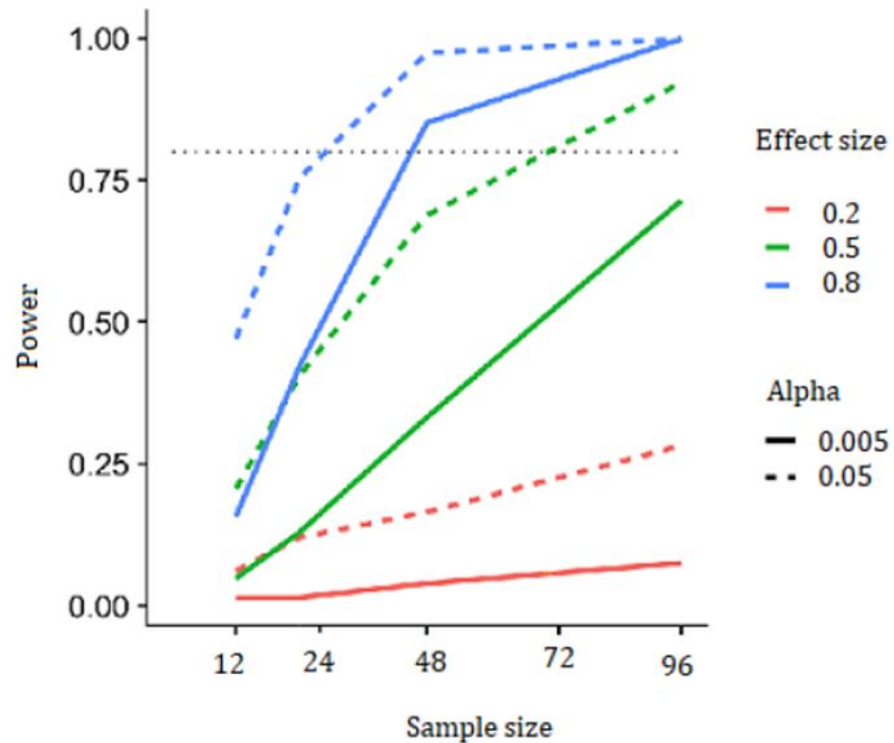
Required n per group across effect sizes



This is why an in vitro study run with 3 vessels per group only reliably detects very large effects ($d \geq \sim 1.6$). Anything subtler needs more vessels or a smarter design (two slides ahead).

The sample size-power trade-off

For fixed α and effect size, power and n move together.



Going from 80% to 90% power requires ~27% more participants.

Going from 80% to 95% requires ~51% more.

Effect size → sample size

Two independent groups, $\alpha = 0.05$ (two-sided), power = 80% .

Cohen's d	Interpretation	n per group	Total n (2 groups)
0.2	small	392	784
0.5	medium	63	126
0.8	large	25	50
1.0	butyrate example	16	32
1.2	very large	11	22
2.0	huge	4	8

A paired, baseline-corrected design cuts your n sharply

Recall the pre-treatment baseline control from earlier, using it turns an unpaired comparison into a paired one.

- This removes the vessel-to-vessel variance that both groups share, shrinking the effective standard deviation of the comparison
- Effective SD of the difference: $\sigma_{diff} = \sigma \times \sqrt{2(1-\rho)}$, where ρ is the baseline-to-endpoint correlation within a vessel
- The sample-size formula for a paired (one-sample) test drops the factor of 2: $n \approx (z_{1-\alpha/2} + z_{1-\beta})^2 / d_{paired}^2$

With $\rho = 0.6$ correlation:

$$\sigma_{diff} = 4.0 \times \sqrt{2 \times 0.4} = 3.58 \text{ mM}$$

$$d_{paired} = 4.0 / 3.58 = 1.12$$

$$n = 7.84 / 1.12^2 \approx 6.3 \rightarrow 7 \text{ vessels}$$

ρ still has to come from real pilot data, not a guess, an optimistic ρ understates the n you actually need.

Web-based calculation tools

PowerAndSampleSize.com

- 120+ calculators backed by peer-reviewed R packages.
- Generates shareable URLs and produces output suitable for manuscripts and grant applications.
- Integrates naturally with the pwr package we use in the R.

A pre-registration checklist

- ☐ Independent units defined: donors, vessels (technical replicates labeled separately).
- ☐ Controls specified: blank, vehicle, positive, baseline.
- ☐ Randomization and blocking plan written before day one.
- ☐ Confounders logged: run, position, operator, batch.
- ☐ Sampling schedule fixed, stabilization period defined.
- ☐ Effect size, α , and power stated explicitly, and n calculated from them, not assumed.

Take home

- Identify every source of variation, donor, vessel, batch, technical, before you run a single vessel
- Your statistical unit is the independent biological replicate; protect that definition
- Randomize and block what you can't eliminate
- Calculate n from a real effect size and your chosen α and power, and reach for a paired design before adding vessels
- Tomorrow: choosing the right test, correcting for multiple comparisons, and reading effect size

Any question?

Homework :)

How would you design a crossover and a longitudinal study in a chemostat system, and how would that change the power calculation?

***Note:** Tomorrow we will discuss it.
You are not allowed to use AI tools ;)*